DBMS TASK

TASK:= SALES AND CUSTOMER ANALYTICS SYSTEM

1. INTRODUCTION

The Sales and Customer Analytics System is a database management project developed using Oracle SQL. It is designed to manage and analyze sales transactions, customer purchase details, product information, and category-wise sales data. The system uses aggregate functions such as COUNT(), SUM(), AVG(), MIN(), and MAX() to generate meaningful sales reports. It also helps identify top customers, best-selling products, and sales performance across different product categories. This project provides useful information for understanding business performance and supporting effective decision-making.

2. OBJECTIVES

1. To apply aggregate functions such as COUNT(), SUM(), AVG(), MIN(), and MAX().
2. To generate total sales reports.
3. To identify top customers based on purchase amount.
4. To identify the best-selling products.
5. To perform category-wise sales analysis.

3. TABLES USED

The following existing tables are used in this project:

1. CUSTOMER
2. PRODUCT
3. CATEGORY
4. ORDERS
5. ORDER_ITEM

4. TABLE STRUCTURE

Table 1: CUSTOMER

Column Name	Data Type	Description
CUSTOMER_ID	NUMBER(38)	Primary Key
FIRST_NAME	VARCHAR2(50)	Customer first name
LAST_NAME	VARCHAR2(50)	Customer last name
EMAIL	VARCHAR2(100)	Customer email

PHONE	VARCHAR2(15)	Contact number
PASSWORD	VARCHAR2(100)	Customer password
ADDRESS	VARCHAR2(255)	Customer address

Table 2: PRODUCT

Column Name	Data Type	Description
PRODUCT_ID	NUMBER	Primary Key
PRODUCT_NAME	VARCHAR2(100)	Product name
CATEGORY_ID	NUMBER	Category reference
PRICE	NUMBER(10,2)	Product price
STOCK	NUMBER	Available stock
SELLER_ID	NUMBER	Seller reference

Table 3: CATEGORY

Column Name	Data Type	Description
CATEGORY_ID	NUMBER	Primary Key
CATEGORY_NAME	VARCHAR2(50)	Category name
DESCRIPTION	VARCHAR2(100)	Category description

Table 4: ORDERS

Column Name	Data Type	Description
ORDER_ID	NUMBER	Primary Key
CUSTOMER_ID	NUMBER	Customer reference
ORDER_DATE	DATE	Order date
DELIVERY_DATE	DATE	Delivery date
ORDER_STATUS	VARCHAR2(20)	Order status
TOTAL_AMOUNT	NUMBER(10,2)	Total order amount

Table 5: ORDER_ITEM

Column Name	Data Type	Description
ORDERITEM_ID	NUMBER	Primary Key
ORDER_ID	NUMBER	Order reference
PRODUCT_ID	NUMBER	Product reference
QUANTITY	NUMBER	Product quantity
UNIT_PRICE	NUMBER(10,2)	Price per unit

5. OPERATIONS PERFORMED

1. Aggregate Functions

Applied COUNT(), SUM(), AVG(), MIN(), and MAX() to calculate the total number of orders, total sales, average sales, minimum sales, and maximum sales.

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
           8

SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Delivered';

TOTAL_SALES
-----------
       7400

SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_SALES
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Delivered';

AVERAGE_SALES
-------------
         1850

SQL> SELECT MIN(TOTAL_AMOUNT) AS MINIMUM_SALES
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Delivered';

MINIMUM_SALES
-------------
         1200

SQL> SELECT MAX(TOTAL_AMOUNT) AS MAXIMUM_SALES
  2  FROM ORDERS
  3  WHERE ORDER_STATUS = 'Delivered';

MAXIMUM_SALES
-------------
         2500
```

2. Total Sales Report

Generated order-wise and status-wise sales reports to analyze total sales and order details.

```
SQL> SELECT
  2      O.ORDER_ID,
  3      O.CUSTOMER_ID,
  4      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      O.TOTAL_AMOUNT
  8  FROM ORDERS O
  9  JOIN CUSTOMER C
 10  ON O.CUSTOMER_ID = C.CUSTOMER_ID
 11  ORDER BY O.ORDER_ID;

  ORDER_ID CUSTOMER_ID
---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT ORDER_STATUS          TOTAL_AMOUNT
--------- -------------------- ------------
       201          101
Arun Kumar
10-JAN-26 Delivered                    1500

       202          102
Priya Sharma
11-JAN-26 Delivered                    2500

  ORDER_ID CUSTOMER_ID
---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT ORDER_STATUS          TOTAL_AMOUNT
--------- -------------------- ------------

       203          103
Rahul Verma
12-JAN-26 Cancelled                    1800

       204          104
Sneha Reddy

  ORDER_ID CUSTOMER_ID
---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT ORDER_STATUS          TOTAL_AMOUNT
--------- -------------------- ------------
13-JAN-26 Pending                      3000

       205          105
Karthik Raj
14-JAN-26 Delivered                    1200

       206          106

  ORDER_ID CUSTOMER_ID
---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT ORDER_STATUS          TOTAL_AMOUNT
--------- -------------------- ------------
Divya Nair
15-JAN-26 Cancelled                    3500
```

3. Top Customer Analysis

Identified customers with the highest purchase amounts using customer and order information.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(O.ORDER_ID) AS TOTAL_ORDERS,
  5      SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  WHERE O.ORDER_STATUS = 'Delivered'
 10  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME
 11  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE
------------ --------------
         102
Priya Sharma
           1            2500

         107
Vignesh Mohan
           1            2200

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE
------------ --------------

         101
Arun Kumar
           1            1500

         105
Karthik Raj

CUSTOMER_ID
-----------
CUSTOMER_NAME
```

4. Best-Selling Product Analysis

Analyzed product-wise quantities sold and total sales amounts to identify the best-selling products.

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(OI.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P
  6  JOIN ORDER_ITEM OI
  7  ON P.PRODUCT_ID = OI.PRODUCT_ID
  8  JOIN ORDERS O
  9  ON OI.ORDER_ID = O.ORDER_ID
 10  WHERE O.ORDER_STATUS = 'Delivered'
 11  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
 12  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       103
Headphones
                  1

       102
Smartphone
                  1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

       105
T-Shirt
                  1

       104
Smart Watch

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
                  1
```

5. Category-Wise Sales Analysis

Calculated total sales and quantities sold for each product category.

```
SQL> SELECT
  2      C.CATEGORY_ID,
  3      C.CATEGORY_NAME,
  4      SUM(OI.QUANTITY * OI.UNIT_PRICE) AS TOTAL_SALES
  5  FROM CATEGORY C
  6  JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  JOIN ORDER_ITEM OI
  9  ON P.PRODUCT_ID = OI.PRODUCT_ID
 10  JOIN ORDERS O
 11  ON OI.ORDER_ID = O.ORDER_ID
 12  WHERE O.ORDER_STATUS = 'Delivered'
 13  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 14  ORDER BY TOTAL_SALES DESC;

CATEGORY_ID CATEGORY_NAME                                          TOTAL_SALES
----------- -------------------------------------------------- -----------
          1 Electronics                                               6200
          2 Clothing                                                  1200
```

6.AGGREGATE FUNCTION RESULTS

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_ORDERS,
  3      SUM(CASE WHEN ORDER_STATUS = 'Delivered' THEN 1 ELSE 0 END) AS DELIVERED_ORDERS,
  4      SUM(CASE WHEN ORDER_STATUS = 'Pending' THEN 1 ELSE 0 END) AS PENDING_ORDERS,
  5      SUM(CASE WHEN ORDER_STATUS = 'Cancelled' THEN 1 ELSE 0 END) AS CANCELLED_ORDERS,
  6      SUM(CASE WHEN ORDER_STATUS = 'Delivered' THEN TOTAL_AMOUNT ELSE 0 END) AS TOTAL_SALES,
  7      AVG(CASE WHEN ORDER_STATUS = 'Delivered' THEN TOTAL_AMOUNT END) AS AVERAGE_SALES
  8  FROM ORDERS;

TOTAL_ORDERS DELIVERED_ORDERS PENDING_ORDERS CANCELLED_ORDERS TOTAL_SALES
------------ ---------------- -------------- ---------------- -----------
AVERAGE_SALES
-------------
           8                4              2                2        7400
        1850
```

7. ORDER STATUS REPORT

```
SQL> SELECT ORDER_STATUS, COUNT(*)
  2  FROM ORDERS
  3  GROUP BY ORDER_STATUS;

ORDER_STATUS          COUNT(*)
-------------------- ----------
Delivered                     4
Cancelled                     2
Pending                       2
```

```
SQL> SELECT ORDER_ID, CUSTOMER_ID, ORDER_STATUS
  2  FROM ORDERS;

  ORDER_ID CUSTOMER_ID ORDER_STATUS
---------- ----------- --------------------
       201         101 Delivered
       202         102 Delivered
       203         103 Cancelled
       204         104 Pending
       205         105 Delivered
       206         106 Cancelled
       207         107 Delivered
       208         101 Pending

8 rows selected.
```

8. CONCLUSION

The Sales and Customer Analytics System was successfully implemented using Oracle SQL. Aggregate functions were used to analyze sales information. Customer purchase reports, best-selling product reports, and category-wise sales reports were generated. This system helps in understanding sales performance and supports better business decision-making.